TUTORIAL 13

Diagonalization, Cayley-Hamilton, Jordan Form, and Inner Products

MOTIVATION AND GUIDING IDEA

Diagonalization replaces coordinates whose components interact by coordinates in which each component is multiplied by its own scalar. This helps compute powers, understand update processes, and identify preserved or decaying components. Finding eigenvalues is only the first step: enough linearly independent eigenvectors must exist over the field being used.

NOTATION AND CONVENTIONS

- $A \in M_n(\mathbb{F})$ is square; $\lambda \in \mathbb{F}$ is a scalar, and $v \in \mathbb{F}^n \setminus \{0\}$ is an eigenvector when $Av = \lambda v$.
- $E_\lambda(A) = \ker(A - \lambda I_n) = \ker(\lambda I_n - A)$ is the eigenspace. The notations V_λ and $N(\lambda I_n - A)$ used in the notes describe the same space.
- $p_A(t) = \det(tI_n - A)$ is the characteristic polynomial; t is neither a vector nor a matrix parameter. The notations $P_A(x)$ and $\chi_A(t)$ express the same convention with a different variable name.
- $r_a(\lambda)$ is the algebraic multiplicity, and $r_g(\lambda) = \dim E_\lambda(A)$ is the geometric multiplicity.
- $P = (v_1 \ \cdots \ v_n)$ diagonalizes A when its columns form an eigenbasis. If $Av_i = \lambda_i v_i$, then $AP = PD$, with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.
- For $f(t) = \sum_{j=0}^{m} c_j t^j$, define $f(A) = \sum_{j=0}^{m} c_j A^j$, where $A^0 = I_n$.

DEFINITION 13.1

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible $P \in M_n(\mathbb{F})$ and a diagonal $D \in M_n(\mathbb{F})$ with $P^{-1}AP = D$. Equivalently, $\mathbb{F}^n$ has a basis of eigenvectors. The precise criterion is that p_A split into linear factors over $\mathbb{F}$ and that $r_g(\lambda) = r_a(\lambda)$ for every eigenvalue. Splitting is automatic over $\mathbb{C}$; over $\mathbb{R}$, it must be checked.

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NOTE

If $A = PDP^{-1}$, then $f(A) = Pf(D)P^{-1}$ for every polynomial f. In particular, $A^k = PD^kP^{-1}$ for every integer $k \geq 0$; negative powers require invertibility. If $Av = \lambda v$ and $v \neq 0$, then $f(A)v = f(\lambda)v$, but distinct eigenvalues of A can give the same value of f. Diagonalizability of a power or polynomial of A therefore does not imply diagonalizability of A.

NOTE

Sum eigenspace dimensions over distinct eigenvalues. Eigenspaces for distinct eigenvalues form a direct sum. Their dimensions sum to n exactly when there are enough eigenvectors for a basis; a smaller sum means that an eigenbasis is missing.

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to identify directions preserved by a linear map, compute eigenspaces and multiplicities, test diagonalizability, and use an eigenbasis to calculate powers and understand an iterative process. We distinguish statements that hold generally from those requiring a particular field or additional hypotheses.

13.1 Eigenspaces, the Characteristic Polynomial, and Multiplicities

DEFINITION 13.2

Definition: characteristic polynomial and multiplicities

1. **Characteristic polynomial $P_A(x)$:** It is defined by
$$P_A(x) = \det(xI_n - A)$$
2. **Algebraic multiplicity $r_a(\lambda)$:** The multiplicity (exponent) of λ as a root of the characteristic polynomial $P_A(x)$.
3. **Geometric multiplicity $r_g(\lambda)$:** The dimension of the eigenspace for λ:
$$r_g(\lambda) = \dim(V_\lambda) = \dim(N(\lambda I_n - A))$$

EXERCISE 13.1

Let
$$A = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$
Please find its characteristic polynomial, eigenspaces, and multiplicities. Please determine whether A is diagonalizable.

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EXERCISE 13.1 CONTINUED

WORKED SOLUTION

Triangularity gives

$$p_A(\lambda) = (\lambda + 2)^2(\lambda - 3)^2.$$

Each eigenvalue $-2, 3$ has algebraic multiplicity 2. The equation $(A + 2I)v = 0$ gives $v_2 = 0$ and $5v_3 = 5v_4 = 0$, so

$$E_{-2} = \text{span}\{e_1\}, \qquad \dim E_{-2} = 1.$$

The equation $(A - 3I)v = 0$ gives $-5v_1 + v_2 = 0$ and $-5v_2 = 0$, so

$$E_3 = \text{span}\{e_3, e_4\}, \qquad \dim E_3 = 2.$$

The eigenspace dimensions sum to $3 < 4$, so A is not diagonalizable.

🏆 **Victory!**

DEFINITION 13.3

For $\lambda \in \mathbb{F}$ and $A \in M_n(\mathbb{F})$, λ is an eigenvalue exactly when $\ker(A - \lambda I_n) \neq \{0\}$, equivalently $p_A(\lambda) = 0$. The eigenspace contains zero, but zero itself is not an eigenvector. For an eigenvalue, the algebraic multiplicity is the integer $r \geq 1$ such that $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$; the geometric multiplicity satisfies $1 \leq r_g(\lambda) \leq r_a(\lambda)$.

13.2 Diagonalization and Eigenbases

MOTIVATION AND GUIDING IDEA

Main objective: find a basis of eigenvectors

DEFINITION 13.4

Definition: diagonalizable matrix and diagonalizing basis

Let $A \in M_n(\mathbb{F})$.

1. A is **diagonalizable over** $\mathbb{F}$ if it is similar over $\mathbb{F}$ to a diagonal matrix. That is, there are an invertible matrix P and a diagonal matrix D, both over $\mathbb{F}$, such that

   $$D = P^{-1}AP \iff A = PDP^{-1}$$

   In this case, the diagonal entries of D are the eigenvalues of A:

   $$D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

2. The columns of the diagonalizing matrix P form a basis of **eigenvectors** of A, in the order of the diagonal entries of D. Since P is invertible, its columns form a basis of $\mathbb{F}^n$. This is a **diagonalizing basis** , or **eigenbasis** .

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PROPOSITION 13.1

Diagonalization theorem (criterion for diagonalizability)
A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ **if and only if** both conditions hold:

1. **Complete splitting:** The characteristic polynomial $P_A(x)$ splits into linear factors over $\mathbb{F}$, meaning that all its roots (counted with multiplicity) lie in $\mathbb{F}$.
2. **Equal multiplicities:** For each eigenvalue λ, the geometric multiplicity equals the algebraic multiplicity:
$$r_g(\lambda) = r_a(\lambda)$$

Useful consequence: If a matrix A of size n has n **distinct** eigenvalues in $\mathbb{F}$, then $r_a = 1$ for each. Consequently $r_g = 1$ (since $1 \le r_g \le r_a$), and A is diagonalizable over $\mathbb{F}$.

NOTE

Why does $1 \le r_g(\lambda) \le r_a(\lambda)$ always hold? Write $r = \dim V_\lambda$. Since λ is an eigenvalue, a nonzero eigenvector exists, so $r \ge 1$. Extend a basis of V_λ to a basis of the full space. In this basis, the matrix has the form
$$\begin{pmatrix} \lambda I_r & C \\ 0 & D \end{pmatrix}.$$
The lower-left block is zero because each of the first r basis vectors maps to a multiple of itself. Thus the characteristic polynomial is $(x-\lambda)^r \det(xI - D)$, and $r_g(\lambda) = r \le r_a(\lambda)$.

DEFINITION 13.5

Definition: a diagonalizable linear operator
Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V of dimension n over $\mathbb{F}$. The operator T is **diagonalizable over** $\mathbb{F}$ if there is a basis $\mathcal{B}$ of V in which $[T]_{\mathcal{B}}$ is diagonal.
Such a basis necessarily consists of eigenvectors of T and is called a **diagonalizing basis** .

PROPOSITION 13.2

Let $v_1, \dots, v_k$ be eigenvectors of a matrix A (or an operator T), associated with **distinct** eigenvalues $\lambda_1, \dots, \lambda_k$.
Then $\{v_1, \dots, v_k\}$ is **linearly independent** .

MOTIVATION AND GUIDING IDEA

An algorithm for diagonalizing a linear operator
To diagonalize T (or find a diagonalizing basis), proceed as follows:

1. Choose a convenient basis $\mathcal{C}$ of V (often the standard basis).
2. Form its representing matrix $A = [T]_{\mathcal{C}}$.
3. Solve the matrix problem:
   - Compute the characteristic polynomial $P_A(x)$.
   - Find the eigenvalues λ in $\mathbb{F}$.

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- For each λ, find a basis of the eigenspace (the kernel) V_λ.

4. Test diagonalizability: add the geometric multiplicities ($\sum r_g(\lambda)$). If the sum is less than n, the operator is not diagonalizable over $\mathbb{F}$. If it equals n, continue.

5. Recover the diagonalizing basis: The vectors found in step 3 are coordinate vectors (in $\mathbb{F}^n$). Convert them back to vectors of V using the initial basis $\mathcal{C}$. The combined lists form the diagonalizing basis $\mathcal{B}$.

6. The representing matrix in the new basis is

$$[T]_{\mathcal{B}} = D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

13.2.1 Exercises on Diagonalization

EXAMPLE 13.1

A polynomial that splits into linear factors over $\mathbb{R}$: $p(x) = (x+2)(x-1)(x-4)$.

EXAMPLE 13.2

A polynomial that **does not** split into linear factors over $\mathbb{R}$: $q(x) = x^2 + 4$.
Observe that $q(x)$ **does** split into linear factors over $\mathbb{C}$: $q(x) = (x-2i)(x+2i)$

EXERCISE 13.2

For each matrix, please determine whether it is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$. Whenever it is diagonalizable, please give suitable P, D over the relevant field:

$$A_1 = \begin{pmatrix} 5 & 3 \\ 0 & 5 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -2 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix}.$$

WORKED SOLUTION

1. For A_1, the characteristic polynomial is $(\lambda - 5)^2$. The equation $(A_1 - 5I)v = 0$ forces $3v_2 = 0$, so the eigenspace is $\text{span}\{(1,0)^T\}$ and has dimension 1. Its algebraic multiplicity 2 exceeds its geometric multiplicity, so it is not diagonalizable over either field.

2. A_2 has distinct eigenvalues $-2, 1, 4$. For -2, choose e_1. For 1, the equation $-3v_1 + 3v_2 = 0$ gives $(1,1,0)^T$. For 4, choose e_3. Hence

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D = \text{diag}(-2, 1, 4).$$

   The columns are linearly independent, and their images give $A_2P = PD$. This is a diagonalization over $\mathbb{R}$, and hence also over $\mathbb{C}$.

3. The characteristic polynomial of A_3 is $\lambda^2 + 4$. It has no real root, so there is no real diagonalization. Over $\mathbb{C}$, the eigenvalues are $2i, -2i$, with respective eigenvectors

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EXERCISE 13.2 CONTINUED

$(2i, 1)^T, (-2i, 1)^T$:

$$A_3(\lambda, 1)^T = (-4, \lambda)^T = \lambda(\lambda, 1)^T \quad \text{when } \lambda^2 = -4.$$

Therefore

$$P_{\mathbb{C}} = \begin{pmatrix} 2i & -2i \\ 1 & 1 \end{pmatrix}, \quad D_{\mathbb{C}} = \operatorname{diag}(2i, -2i).$$

Since $\det P_{\mathbb{C}} = 4i \neq 0$, this is a valid complex diagonalization.

And we're done. Victory!

EXERCISE 13.3

As in the earlier transpose exercise, assume $n \geq 2$ and consider $T : M_n(\mathbb{R}) \to M_n(\mathbb{R})$ defined by

$$T(A) = pA + qA^T$$

where $p, q \in \mathbb{R}$ and $q \neq 0$.
Please find the characteristic polynomial $P_T(x)$ and the algebraic and geometric multiplicities (r_a, r_g) of each eigenvalue found earlier.

WORKED SOLUTION

Previously we decomposed the full space $V = M_n(\mathbb{R})$ into a direct sum of eigenspaces:

$$V = V_{p+q} \oplus V_{p-q} = S \oplus K$$

Here S is the space of symmetric matrices and K is the space of skew-symmetric matrices.

1. **For the eigenvalue $\lambda_1 = p + q$:** The eigenspace consists of the symmetric matrices:

   $$V_{\lambda_1} = S = \{A \in M_n(\mathbb{R}) \mid A^T = A\}$$

   Its geometric multiplicity is its dimension (the number of entries on or above the diagonal):

   $$r_g(p + q) = \dim(S) = \frac{n(n + 1)}{2}$$

2. **For the eigenvalue $\lambda_2 = p - q$:** The eigenspace consists of the skew-symmetric matrices:

   $$V_{\lambda_2} = K = \{A \in M_n(\mathbb{R}) \mid A^T = -A\}$$

   Its geometric multiplicity is its dimension (the number of entries strictly above the diagonal, since diagonal entries must be zero):

   $$r_g(p - q) = \dim(K) = \frac{n(n - 1)}{2}$$

3. **The characteristic polynomial $P_T(x)$:** The geometric multiplicities sum to the dimension of the full space (n^2), so the operator is diagonalizable. Thus each algebraic multiplicity equals its geometric multiplicity ($r_a = r_g$).

   The characteristic polynomial is the product of the linear factors, each raised to its al-

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EXERCISE 13.3 CONTINUED

gebraic multiplicity:

$$P_T(x) = (x - \lambda_1)^{r_a(\lambda_1)} \cdot (x - \lambda_2)^{r_a(\lambda_2)}$$

Substitute the values found above:

$$P_T(x) = \Big(x - (p+q)\Big)^{\frac{n(n+1)}{2}} \cdot \Big(x - (p-q)\Big)^{\frac{n(n-1)}{2}}$$

You know what that means? V-I-C-T-O-R-Y!

NOTE

The dimension of a solution subspace can be found by counting independent homogeneous linear constraints: If L is the linear constraint map on V, then

$$\dim \ker L = \dim V - \operatorname{rank} L$$

Here the ambient space is $M_n(\mathbb{R})$, of dimension n^2.

1. **For symmetric matrices (S):** The condition is $A_{ij} = A_{ji}$. This is a constraint only for $i < j$ (on the diagonal $A_{ii} = A_{ii}$ is an identity, and the equations below the diagonal repeat those above it). There are $\frac{n(n-1)}{2}$ independent constraints, one per entry strictly above the diagonal. Thus the dimension is

$$n^2 - \frac{n(n-1)}{2} = \frac{2n^2 - n^2 + n}{2} = \frac{n(n+1)}{2}$$

2. **For skew-symmetric matrices (K):** The condition is $A_{ji} = -A_{ij}$. There are two types of constraints:

   - For $i < j$, there are $\frac{n(n-1)}{2}$ constraints (as in the symmetric case).
   - For $i = j$ (on the diagonal), $A_{ii} = -A_{ii}$ implies $A_{ii} = 0$. There are n such constraints.

   The total number of independent constraints is $\frac{n(n-1)}{2} + n$. Thus the dimension is

$$n^2 - \left(\frac{n(n-1)}{2} + n\right) = \frac{n(n-1)}{2}$$

NOTE

A useful check: the sum of the exponents (degrees) in the characteristic polynomial must equal the dimension of the space on which the operator acts.

$$\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n^2 + n + n^2 - n}{2} = \frac{2n^2}{2} = n^2$$

This is indeed the dimension of $M_n(\mathbb{R})$.

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EXERCISE 13.4

Let $n \in \mathbb{N}$, and consider a general upper triangular matrix $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Its diagonal entries $\lambda_1, \ldots, \lambda_n$ are **pairwise distinct** (that is, $\lambda_i \neq \lambda_j$ whenever $i \neq j$).
Define the polynomial $P(x) = 7x^7 - 4x^4$.
Please prove that there **exists** a matrix $X \in M_n(\mathbb{R})$ satisfying

$$P(X) = A$$

WORKED SOLUTION

Solution strategy: Although A has unspecified entries (asterisks) above the diagonal, the essential fact is that its diagonal entries are **distinct**. A matrix of size n with n distinct eigenvalues is diagonalizable. We can therefore work with the simpler diagonal matrix, solve the resulting scalar problem using calculus, and return to the original coordinates.

1. The characteristic polynomial of an upper triangular matrix is the product of its diagonal factors:
$$P_A(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n)$$
Its eigenvalues are therefore exactly $\lambda_1, \ldots, \lambda_n$. They are distinct, so A has n linearly independent eigenvectors. Thus A is diagonalizable: there are an invertible M and a diagonal matrix $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ such that
$$A = MDM^{-1}$$

2. Seek a solution of the form $X = MEM^{-1}$, where $E = \mathrm{diag}(c_1, \ldots, c_n)$ is a diagonal matrix to be constructed. The requirement $P(X) = A$ becomes the following equation for diagonal matrices:
$$P(E) = D$$
Since both are diagonal, this is equivalent to n separate scalar equations:
$$P(c_i) = \lambda_i \quad \text{for every } i = 1, \ldots, n$$

3. Consider the real function $f(t) = 7t^7 - 4t^4$.
   - It is continuous.
   - Its leading term has odd degree and positive coefficient, so $\lim_{t \to \pm\infty} f(t) = \pm\infty$.

   The intermediate value theorem implies that f maps onto $\mathbb{R}$. Thus for each eigenvalue λ_i, regardless of the unspecified entries in A, there exists $c_i \in \mathbb{R}$ such that $f(c_i) = \lambda_i$.

4. **Construct the solution:** Choose these c_i as the diagonal entries of E. Define
$$X = MEM^{-1}$$

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EXERCISE 13.4 CONTINUED

and verify:

$$P(X) = MP(E)M^{-1} = M\,\mathrm{diag}(P(c_1),\dots,P(c_n))M^{-1}$$
$$= M\,\mathrm{diag}(\lambda_1,\dots,\lambda_n)M^{-1} = MDM^{-1} = A$$

This proves existence of the required matrix.

Another one down. Victory!

PROPOSITION 13.3

If $A \sim B$, the following quantities and properties are preserved:

1. **Determinant:** $\det(A) = \det(B)$.
2. **Trace:** $\mathrm{tr}(A) = \mathrm{tr}(B)$.
3. **Characteristic polynomial:** $P_A(x) = P_B(x)$.
4. **Eigenvalues:** A and B have exactly the same eigenvalues, with the same algebraic and geometric multiplicities.
5. **Rank:** $\mathrm{rank}(A) = \mathrm{rank}(B)$.
6. **Diagonalizability:** A is diagonalizable $\iff$ B is diagonalizable.

NOTE

Warning:
Equality of characteristic polynomials (and hence of eigenvalues, determinants, and traces) **does not guarantee** similarity.

A standard counterexample is $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. Both have characteristic polynomial x^2, but they are not similar: the zero matrix is similar only to itself (prove that the same is true of every scalar matrix).

13.3 The Cayley-Hamilton Theorem

DEFINITION 13.6

For $f(t) = \sum_{k=0}^{m} a_k t^k$ and $A \in M_n(\mathbb{F})$ with $n \geq 1$, define

$$f(A) = \sum_{k=0}^{m} a_k A^k, \qquad A^0 = I_n.$$

In particular, the constant term a_0 becomes $a_0 I_n$. Our convention is

$$\chi_A(t) = \det(tI_n - A) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0.$$

The convention $\det(A - tI_n)$ gives $(-1)^n \chi_A(t)$ and hence the same vanishing identity.

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PROPOSITION 13.4

Every square matrix satisfies its characteristic polynomial:

$$\chi_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

We prove this using the adjugate. For $n = 1$, the identity is $A - aI_1 = 0$ with $A = (a)$. For $n \geq 2$, write

$$B(t) = \operatorname{adj}(tI_n - A) = \sum_{j=0}^{n-1} B_j t^j.$$

Every cofactor is a polynomial of degree at most $n-1$. The adjugate identity gives

$$(tI_n - A)B(t) = \chi_A(t) I_n.$$

Comparing coefficients of these polynomials in t yields

$$B_{n-1} = I_n, \qquad B_{j-1} - AB_j = c_j I_n \quad (1 \leq j \leq n-1), \qquad -AB_0 = c_0 I_n.$$

Starting at the top coefficient, use $B_{j-1} = AB_j + c_j I_n$. First $B_{n-2} = A + c_{n-1}I_n$, and continued substitution gives

$$B_0 = A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n.$$

Substituting this expression into $-AB_0 = c_0 I_n$ and rearranging proves the theorem.

RECALL AND TOOLS

Polynomial division by χ_A gives a computational tool. If

$$f(t) = q(t)\chi_A(t) + r(t), \qquad \deg r < n,$$

where $r = 0$ is also allowed, then

$$f(A) = q(A)\chi_A(A) + r(A) = r(A).$$

Evaluation at the same matrix preserves multiplication: in the expansion of $q(A)g(A)$, each product of powers is $A^i A^j = A^{i+j}$. Thus high powers reduce to combinations of $I, A, \ldots, A^{n-1}$. If A is invertible, then $c_0 = (-1)^n \det A \neq 0$. Put $c_n = 1$. Multiplying the Cayley-Hamilton identity by A^{-1} gives

$$A^{-1} = -\frac{1}{c_0} \sum_{k=1}^{n} c_k A^{k-1}.$$

The same statements hold for a linear map $T : V \to V$ on a finite-dimensional space, with I_V in place of I_n.

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EXERCISE 13.5

Consider the real matrix

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Please find χ_A, and please use Cayley-Hamilton to compute A^{-1} and A^5.

WORKED SOLUTION

Expansion along the first row gives

$$\begin{aligned} \chi_A(t) &= \det \begin{pmatrix} t & 0 & -1 \\ -1 & t & 1 \\ 0 & -1 & t-2 \end{pmatrix} \\ &= t\big(t(t-2)+1\big) - 1 = t^3 - 2t^2 + t - 1. \end{aligned}$$

Hence

$$A^3 = 2A^2 - A + I.$$

It follows that $A(A^2 - 2A + I) = I$ and $(A^2 - 2A + I)A = I$, so $A^{-1} = A^2 - 2A + I$. Direct calculation yields

$$A^2 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & -1 & -1 \\ 1 & 2 & 3 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ -2 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Reduce the next powers in turn:

$$\begin{aligned} A^4 &= 2A^3 - A^2 + A = 2(2A^2 - A + I) - A^2 + A = 3A^2 - A + 2I, \\ A^5 &= 3A^3 - A^2 + 2A = 3(2A^2 - A + I) - A^2 + 2A = 5A^2 - A + 3I. \end{aligned}$$

Finally,

$$A^5 = \begin{pmatrix} 3 & 5 & 9 \\ -1 & -2 & -4 \\ 5 & 9 & 16 \end{pmatrix}.$$

We've got it. Victory!

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EXERCISE 13.6

Consider the real matrix

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{pmatrix}.$$

Please find a formula for A^m for every integer $m \geq 0$, and please determine whether A is diagonalizable.

WORKED SOLUTION

Its characteristic polynomial is $(t-3)^3$. Put $N = A - 3I$. Cayley-Hamilton gives $N^3 = 0$. In this case this can also be checked directly:

$$N = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

The matrices $3I$ and N commute, so the binomial theorem applies to $(3I+N)^m$. Every term containing a power of N greater than 2 vanishes. With $\binom{m}{2} = m(m-1)/2$, we obtain

$$A^m = 3^m I + m3^{m-1}N + \binom{m}{2}3^{m-2}N^2 = \begin{pmatrix} 3^m & 2m3^{m-1} & 2m(m-1)3^{m-2} \\ 0 & 3^m & 2m3^{m-1} \\ 0 & 0 & 3^m \end{pmatrix}.$$

This also holds at $m = 0, 1$: coefficients of the terms absent from the binomial expansion are zero. The only eigenvalue is 3. If A were diagonalizable, its diagonal matrix would be $3I$, implying $A = P(3I)P^{-1} = 3I$. But entry $(1,2)$ of A is 2, so A is not diagonalizable.

That does it. Victory!

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EXERCISE 13.7

Let $A \in M_4(\mathbb{R})$ have

$$\chi_A(t) = t^4 + 2t^3 - 3t^2 - 4t + 5.$$

Please prove that $A + I$ is invertible, and please express its inverse as a polynomial in A of degree at most 3.

WORKED SOLUTION

Division by $t + 1$ gives

$$t^4 + 2t^3 - 3t^2 - 4t + 5 = (t + 1)(t^3 + t^2 - 4t) + 5.$$

Expanding the product gives $t^4 + 2t^3 - 3t^2 - 4t$; the remainder is $5 = \chi_A(-1)$. By Cayley-Hamilton,

$$0 = (A + I)(A^3 + A^2 - 4A) + 5I.$$

Set

$$B = -\frac{1}{5}(A^3 + A^2 - 4A).$$

Then $(A+I)B = I$. Since both matrices are polynomials in A, they commute, so $B(A+I) = I$ as well. Therefore

$$(A + I)^{-1} = -\frac{1}{5}(A^3 + A^2 - 4A).$$

All wrapped up. Victory!

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EXERCISE 13.8

Let $A \in M_3(\mathbb{R})$ have $\chi_A(t) = t(t-2)(t+1)$. Define

$$P = \frac{A^2 + A}{6}, \qquad Q = \frac{A^2 - 2A}{3}, \qquad R = I - P - Q.$$

Please prove that P, Q, R are idempotent (each satisfies $X^2 = X$), and that the product of any two distinct members is zero. Please compute A^m using them for every $m \geq 0$, and please describe the resulting decomposition of each vector into three eigenspaces.

WORKED SOLUTION

Since $\chi_A(t) = t^3 - t^2 - 2t$, Cayley-Hamilton gives

$$A^3 = A^2 + 2A, \qquad A^4 = 3A^2 + 2A.$$

Thus

$$P^2 = \frac{A^4 + 2A^3 + A^2}{36} = \frac{6A^2 + 6A}{36} = P,$$

$$Q^2 = \frac{A^4 - 4A^3 + 4A^2}{9} = \frac{3A^2 - 6A}{9} = Q,$$

$$PQ = QP = \frac{A^4 - A^3 - 2A^2}{18} = 0.$$

Now $RP = PR = P - P^2 - PQ = 0$, and similarly $RQ = QR = 0$. Also,

$$R^2 = (I - P - Q)^2 = I - 2P - 2Q + P^2 + Q^2 + PQ + QP = R.$$

The formulas for P, Q give $A = 2P - Q$. For $m \geq 1$, all mixed products vanish, so

$$A^m = 2^m P + (-1)^m Q.$$

At $m = 0$, we have $A^0 = I = R + P + Q$. To identify the components, calculate

$$AP = \frac{A^3 + A^2}{6} = 2P, \qquad AQ = \frac{A^3 - 2A^2}{3} = -Q, \qquad AR = A - 2P + Q = 0.$$

Consequently, for every $x \in \mathbb{R}^3$,

$$x = Rx + Px + Qx, \qquad Rx \in \ker A, \quad Px \in \ker(A - 2I), \quad Qx \in \ker(A + I).$$

Evaluating the defining polynomials of P, Q, R at $0, 2, -1$ also gives

$$(Px, Qx, Rx) = \begin{cases} (0, 0, x), & Ax = 0, \\ (x, 0, 0), & Ax = 2x, \\ (0, x, 0), & Ax = -x. \end{cases}$$

The decomposition is unique: applying R, P, Q to such a decomposition recovers its three components, since mixed products are zero and each matrix is idempotent.

We've reached the finish. Victory!

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EXERCISE 13.9

Let $A, B \in M_2(\mathbb{C})$ satisfy $\operatorname{tr} A = \operatorname{tr} B = 0$.

1. Please prove that $AB + BA = \operatorname{tr}(AB) I_2$.
2. If also $\det A = -2$, $\det B = 5$, and $\operatorname{tr}(AB) = 0$, please compute $(A+B)^8$.

WORKED SOLUTION

1. The characteristic polynomial of a 2×2 matrix is

$$\chi_X(t) = t^2 - (\operatorname{tr} X)t + \det X.$$

Since $A + B$ also has trace zero, Cayley-Hamilton gives

$$A^2 = -(\det A) I_2, \quad B^2 = -(\det B) I_2, \quad (A+B)^2 = -\det(A+B) I_2.$$

Expanding $(A+B)^2 = A^2 + AB + BA + B^2$ yields

$$AB + BA = cI_2, \qquad c = \det A + \det B - \det(A+B).$$

We have $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ because

$$\operatorname{tr}(AB) = \sum_{i,j=1}^{2} a_{ij} b_{ji} = \sum_{j,i=1}^{2} b_{ji} a_{ij} = \operatorname{tr}(BA).$$

Taking traces gives $2\operatorname{tr}(AB) = 2c$. Division by 2 is valid over $\mathbb{C}$, so $c = \operatorname{tr}(AB)$.

2. We have $A^2 = 2I_2$, $B^2 = -5I_2$, and $AB + BA = 0$. Hence

$$(A+B)^2 = 2I_2 - 5I_2 = -3I_2, \qquad (A+B)^8 = \big((A+B)^2\big)^4 = (-3)^4 I_2 = 81 I_2.$$

Yes! Victory!

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13.4 Diagonalization, Powers, and Parameters

EXERCISE 13.10

Let $A \in M_{5\times 5}(\mathbb{R})$ have 1 on the superdiagonal and 0 elsewhere (that is, $A_{12} = A_{23} = A_{34} = A_{45} = 1$, with all other entries zero).

1. Please show that A is nilpotent, meaning that some positive power is zero, and please find its characteristic polynomial.
2. Please find $\dim \ker(A^2)$ and $\dim \ker(A)$.
3. Please prove that A is not diagonalizable.

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EXERCISE 13.10 CONTINUED

WORKED SOLUTION

1. The matrix is upper triangular with zero diagonal, so $P_A(x) = x^5$. By the Cayley-Hamilton theorem, $A^5 = 0$, and hence A is nilpotent.
2. For A, there is one free variable (the first column is zero and the remaining columns are linearly independent), so $\dim\ker(A) = 1$. In A^2, the entries equal to 1 lie on the second superdiagonal. There are two zero columns and three linearly independent nonzero columns, giving two free variables: $\dim\ker(A^2) = 2$.
3. The only eigenvalue is $\lambda = 0$, of algebraic multiplicity $r_a = 5$. The previous part gives geometric multiplicity $r_g = \dim\ker(A) = 1$. Since $1 \neq 5$, the matrix is not diagonalizable.

V-I-C-T-O-R-Y!

EXERCISE 13.11

Let $A \in M_{n\times n}(\mathbb{F})$, where $\mathbb{F}$ has characteristic different from 2 (so $1_{\mathbb{F}} + 1_{\mathbb{F}} \neq 0_{\mathbb{F}}$, and $2^{-1} = \frac{1}{2}$ exists). Suppose $A^2 = I$.

1. Please prove that every eigenvalue of A lies in $\{1, -1\}$.
2. Please prove that A is diagonalizable. (Hint: express every vector as a sum of vectors in the two eigenspaces.)
3. In this part, assume additionally that $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Given $\mathrm{tr}(A) = t$, please determine $\dim\ker(A - I)$ and $\dim\ker(A + I)$ in terms of n and t.

WORKED SOLUTION

1. Let $\lambda \in \mathbb{F}$ be an eigenvalue. By definition, there is a nonzero vector $v \in \mathbb{F}^n$, $v \neq 0$, with

$$Av = \lambda v$$

Multiply the equality on the left by A:

$$A(Av) = A(\lambda v)$$

Use associativity on the left and take the scalar outside on the right (linearity):

$$A^2 v = \lambda(Av)$$

Substitute $A^2 = I$ on the left and $Av = \lambda v$ on the right:

$$Iv = \lambda(\lambda v)$$

$$v = \lambda^2 v$$

Move all terms to one side:

$$\lambda^2 v - v = 0 \implies (\lambda^2 - 1)v = 0$$

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EXERCISE 13.11 CONTINUED

Since $v \neq 0$ (an eigenvector is nonzero), its scalar coefficient must vanish:

$$\lambda^2 - 1 = 0 \implies \lambda^2 = 1$$

The solutions in the field are $\lambda = 1$ or $\lambda = -1$.

2. To prove diagonalizability, it suffices to find a basis of eigenvectors for the full space. Equivalently, the eigenspaces must span the space: $V = V_1 + V_{-1}$. For any $v \in \mathbb{F}^n$, use the identity below (division by 2 is permitted because the characteristic is not 2):

$$v = \frac{1}{2}(v + Av) + \frac{1}{2}(v - Av)$$

Denote the two summands by

$$u_1 = \frac{1}{2}(v + Av), \quad u_{-1} = \frac{1}{2}(v - Av)$$

Then $v = u_1 + u_{-1}$. Each summand lies in its corresponding eigenspace; a summand may be zero, in which case it is not itself an eigenvector: **Check u_1:** Apply A:

$$A(u_1) = A\left(\frac{1}{2}(v + Av)\right) = \frac{1}{2}(Av + A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av + v) = u_1$$

Thus $u_1 \in \ker(A - I)$ (and if $u_1 \neq 0$, it is an eigenvector for 1). **Check u_{-1}:** Apply A:

$$A(u_{-1}) = A\left(\frac{1}{2}(v - Av)\right) = \frac{1}{2}(Av - A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av - v) = -\frac{1}{2}(v - Av) = -u_{-1}$$

Thus $u_{-1} \in \ker(A + I)$ (and if $u_{-1} \neq 0$, it is an eigenvector for -1). **Conclusion:** Every vector lies in $\ker(A-I)+\ker(A+I)$. Their intersection is trivial: if $Az = z$ and $Az = -z$, then $2z = 0$, so $z = 0$. Consequently, this direct sum is the entire space. Concatenating bases of the two kernels gives an eigenbasis, so A is diagonalizable.

3. The matrix A is diagonalizable, so it is similar to a diagonal matrix D whose diagonal entries are 1 and -1. Write $k = \dim\ker(A - I)$ (the geometric multiplicity of 1, equal to its number of occurrences on the diagonal). $m = \dim\ker(A + I)$ (the geometric multiplicity of -1, equal to its number of occurrences on the diagonal).

Two equations are available: 1. The matrix has size n, so the multiplicities sum to n:

$$k + m = n$$

2. Trace is preserved under similarity, so it equals the sum of the diagonal entries of D:

$$k \cdot (1) + m \cdot (-1) = t \implies k - m = t$$

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EXERCISE 13.11 CONTINUED

This is a system of two equations in the two unknowns k, m. Add the equations:

$$(k+m)+(k-m)=n+t \implies 2k=n+t \implies k=\frac{n+t}{2}$$

Subtract the second equation from the first:

$$(k+m)-(k-m)=n-t \implies 2m=n-t \implies m=\frac{n-t}{2}$$

Answer:

$$\dim\ker(A-I)=\frac{n+t}{2}, \quad \dim\ker(A+I)=\frac{n-t}{2}.$$

The field assumption in this part is essential for interpreting the trace as the difference of integer dimensions. In positive characteristic, the trace determines only their difference as a field element, which need not determine the integer dimensions.

Victory!

EXERCISE 13.12

For $t \in \mathbb{R}$, let

$$A(t)=\begin{pmatrix}3 & t+1 & 0\\ 0 & 3 & 0\\ 0 & 0 & 7\end{pmatrix}.$$

Please find its eigenvalues and multiplicities, please determine when it is diagonalizable, and please prove a formula for $A(t)^n$ for every $n \geq 1$.

WORKED SOLUTION

Its characteristic polynomial is $p_A(\lambda)=(\lambda-3)^2(\lambda-7)$, with algebraic multiplicities $2, 1$. The equation $(A-3I)v=0$ is $(t+1)v_2=0$, $4v_3=0$. Thus the eigenspace for 3 has dimension 2 when $t=-1$ and dimension 1 otherwise. The eigenspace for 7 has dimension 1. Hence $A(t)$ is diagonalizable exactly when $t=-1$. For every $n \geq 1$,

$$A(t)^n=\begin{pmatrix}3^n & n3^{n-1}(t+1) & 0\\ 0 & 3^n & 0\\ 0 & 0 & 7^n\end{pmatrix}.$$

The base case $n=1$ gives $A(t)$. On multiplication by $A(t)$ on the right, entry $(1,2)$ becomes

$$3^n(t+1)+n3^{n-1}(t+1)\cdot 3=(n+1)3^n(t+1),$$

and the diagonal becomes $3^{n+1}, 3^{n+1}, 7^{n+1}$. This proves the induction step. Separately, $A^0=I_3$.

And we're done. Victory!

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EXERCISE 13.13

Let $A \in M_{n\times n}(\mathbb{F})$ be square over $\mathbb{F}$, and let $u, v \in \mathbb{F}^n$ be nonzero column vectors. Define $B = A + uv^T$ (this is a rank-one update).

1. Suppose $x \in \mathbb{F}^n$, $x \neq 0$, satisfies:
   - x is an eigenvector of A with eigenvalue λ (that is, $Ax = \lambda x$).
   - $v^T x = 0$.

   Please prove that x is also an eigenvector of B, and please identify its eigenvalue.

2. Now assume $A = \operatorname{diag}(\alpha_1, \ldots, \alpha_n)$ and $u = v = (1, 1, \ldots, 1)^T$. Please prove that for every scalar λ outside $\{\alpha_1, \ldots, \alpha_n\}$,

$$\lambda \text{ is an eigenvalue of } B \iff 1 + \sum_{i=1}^{n} \frac{1}{\alpha_i - \lambda} = 0$$

3. For $n = 3$ and $A = \operatorname{diag}(-1, 2, 4)$, please compute the characteristic polynomial of B explicitly (with u, v as in the preceding part).

WORKED SOLUTION

1. Check directly by multiplying B by x:

$$Bx = (A + uv^T)x$$

   Expand:

$$= Ax + u(v^T x)$$

   Substitute $Ax = \lambda x$ (because x is an eigenvector of A) and $v^T x = 0$ (by assumption):

$$= \lambda x + u(0) = \lambda x$$

   Conclusion: We obtained $Bx = \lambda x$. Since $x \neq 0$, it is an eigenvector of B with the same eigenvalue λ.

2. We seek the condition $\det(\lambda I - B) = 0$. Substitute B:

$$\det(\lambda I - (A + uv^T)) = \det((\lambda I - A) - uv^T)$$

   Write $D = \lambda I - A$. This is diagonal, with entries $\lambda - \alpha_i$. **Why is D invertible?** By assumption, λ differs from every α_i, so all its diagonal entries are nonzero.

   Factor out D inside the determinant:

$$\det(D - uv^T) = \det(D(I - D^{-1}uv^T))$$

   By multiplicativity ($\det(XY) = \det(X)\det(Y)$),

$$= \det(D) \cdot \det(I - (D^{-1}u)v^T)$$

   For λ to be an eigenvalue, the full expression must be zero. Since D is invertible,

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EXERCISE 13.13 CONTINUED

$\det(D) \neq 0$, so the second factor must vanish:

$$\det(I - (D^{-1}u)v^T) = 0$$

We use $\det(I + xy^T) = 1 + y^T x$, justifying it even when $y^T x = 0$. If $x = 0$, both sides equal 1. Otherwise, extend x to a basis. In this basis, the image of xy^T lies in $\text{span}\{x\}$ and its first column is $(y^T x, 0, \ldots, 0)^T$. Thus $I + xy^T$ has an upper triangular representing matrix with diagonal $(1 + y^T x, 1, \ldots, 1)$, giving determinant $1 + y^T x$. No nonzero eigenvalue of xy^T needs to be assumed. Here $x = -D^{-1}u$ and $y = v$. Hence the determinant is

$$1 - v^T(D^{-1}u) = 0$$

Compute $v^T D^{-1}u$: The diagonal entries of D^{-1} are $\frac{1}{\lambda - \alpha_i}$. Multiplication by the all-ones vectors adds them:

$$\sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 1$$

Rearranging gives

$$1 - \sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 0 \implies 1 + \sum_{i=1}^{n} \frac{1}{\alpha_i - \lambda} = 0$$

3. Here $A = \text{diag}(-1, 2, 4)$, so

$$B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 5 \end{pmatrix}.$$

The preceding identity gives

$$\begin{aligned} p_B(\lambda) &= (\lambda + 1)(\lambda - 2)(\lambda - 4) \\ &\quad - (\lambda - 2)(\lambda - 4) - (\lambda + 1)(\lambda - 4) - (\lambda + 1)(\lambda - 2) \\ &= \lambda^3 - 8\lambda^2 + 12\lambda + 6. \end{aligned}$$

After clearing denominators this is a polynomial identity, valid also at $\lambda = -1, 2, 4$.

You know what that means? V-I-C-T-O-R-Y!

EXERCISE 13.14

Two units exchange information at each step according to

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = A \begin{pmatrix} x_k \\ y_k \end{pmatrix}, \qquad A = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}.$$

Please find an eigenbasis, an explicit formula for A^k and for the two readings, and their limits as $k \to \infty$.

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EXERCISE 13.14 CONTINUED

WORKED SOLUTION

Compute the action of A on two directions:

$$A\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}1\\1\end{pmatrix},\qquad A\begin{pmatrix}1\\-1\end{pmatrix}=\begin{pmatrix}1/3\\-1/3\end{pmatrix}=\frac{1}{3}\begin{pmatrix}1\\-1\end{pmatrix}.$$

These vectors are linearly independent, so

$$P=\begin{pmatrix}1&1\\1&-1\end{pmatrix},\qquad P^{-1}=\frac{1}{2}\begin{pmatrix}1&1\\1&-1\end{pmatrix},\qquad D=\operatorname{diag}(1,1/3).$$

For every integer $k\geq 0$,

$$A^k=PD^kP^{-1}=\frac{1}{2}\begin{pmatrix}1+3^{-k}&1-3^{-k}\\1-3^{-k}&1+3^{-k}\end{pmatrix}.$$

Consequently,

$$x_k=\frac{x_0+y_0}{2}+3^{-k}\frac{x_0-y_0}{2},\qquad y_k=\frac{x_0+y_0}{2}-3^{-k}\frac{x_0-y_0}{2}.$$

The mean is preserved, while the difference $x_k-y_k=3^{-k}(x_0-y_0)$ is divided by three at each step. Both readings therefore converge to the initial mean $(x_0+y_0)/2$. The eigenvalue 1 describes the preserved component, and $1/3$ describes the decaying component.

Another one down. Victory!

EXERCISE 13.15 EXAM QUESTION

Let $\gamma\in\mathbb{Z}_{\geq 0}$. On $\mathbb{C}[x]_{\leq 3}$ define

$$T_\gamma(a_0+a_1x+a_2x^2+a_3x^3)=2a_1-2a_0x+(\gamma a_2-a_3)x^2+3a_3x^3.$$

Please find its matrix in $(1,x,x^2,x^3)$, please determine when it is diagonalizable, and, for the smallest admissible γ, please find an eigenbasis and the characteristic polynomial.

WORKED SOLUTION

The basis images are $-2x, 2, \gamma x^2, -x^2+3x^3$, so

$$A_\gamma=\begin{pmatrix}0&2&0&0\\-2&0&0&0\\0&0&\gamma&-1\\0&0&0&3\end{pmatrix},\qquad p_{T_\gamma}(t)=(t^2+4)(t-\gamma)(t-3).$$

If $\gamma\neq 3$, the four eigenvalues $2i,-2i,\gamma,3$ are distinct, so the map is diagonalizable. At $\gamma=3$, the eigenspace for 3 is $\operatorname{span}\{x^2\}$: the first block has no eigenvalue 3, and the second block forces $a_3=0$. The geometric multiplicity is 1 and the algebraic multiplicity is 2, so diagonalization fails. The smallest admissible parameter is $\gamma=0$. An eigenbasis is

$$\mathcal{D}=(1+ix,1-ix,x^2,3x^3-x^2),\qquad [T_0]_{\mathcal{D}}=\operatorname{diag}(2i,-2i,0,3).$$

For example, $T_0(3x^3-x^2)=-3x^2+9x^3=3(3x^3-x^2)$; the other checks are direct. The

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EXERCISE 13.15 EXAM QUESTION CONTINUED

characteristic polynomial is $(t^2+4)t(t-3)$.

We've got it. Victory!

EXERCISE 13.16 EXAM QUESTION

In $\mathbb{R}^4$, define

$$\mathcal{F} = (e_1, e_1+e_2, e_2+e_3, 2e_3+e_4),$$

and let T send the standard basis $\mathcal{E}$ to this basis in order. Let S satisfy

$$[S]_{\mathcal{F}} = J = \begin{pmatrix} 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 4 & 0 \end{pmatrix}, \qquad U = T^{-1}ST.$$

Please compute $[U]_{\mathcal{E}}$ and please determine whether U is diagonalizable over $\mathbb{R}$ and $\mathbb{C}$. Whenever possible, please find an eigenbasis. For $W = 3I$, please determine whether U, W can have the same matrix in one basis, or whether a change of basis can turn the matrix of U into $3I$.

WORKED SOLUTION

Set

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

We have $[T]_{\mathcal{E}} = P$, $[S]_{\mathcal{E}} = PJP^{-1}$, hence

$$[U]_{\mathcal{E}} = P^{-1}(PJP^{-1})P = J.$$

Its characteristic polynomial is $(t^2+4)(t^2+16)$, so it has no real eigenvalues and no real diagonalization. Over $\mathbb{C}$, an eigenbasis is

$$(e_1 - ie_2, e_1 + ie_2, e_3 - ie_4, e_3 + ie_4),$$

with diagonal matrix $\operatorname{diag}(2i, -2i, 4i, -4i)$. If two maps have the same matrix in the same basis, they are equal. Here $U \neq W$, since $Ue_1 = 2e_2 \neq 3e_1$. A change of basis cannot turn J into $3I$ either: a scalar matrix is similar only to itself.

That does it. Victory!

EXERCISE 13.17 EXAM QUESTION

Let $A \in M_n(\mathbb{R})$ be diagonalizable, with $n \geq 3$, and suppose it has exactly two distinct eigenvalues. Assume

$$\operatorname{rank}(A+I) = n-2, \qquad \operatorname{tr} A = 3n-8, \qquad \det A = 3^{n-2}.$$

Please find the eigenvalues and multiplicities, please write the characteristic polynomial, and

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EXERCISE 13.17 EXAM QUESTION CONTINUED

please compute $\mathrm{tr}(A^k)$ for every $k \geq 1$.

WORKED SOLUTION

Rank-nullity gives $\dim \ker(A+I) = 2$. Thus -1 has geometric multiplicity 2, and diagonalizability makes its algebraic multiplicity 2 as well. The other eigenvalue λ has multiplicity $n-2$. The trace condition gives

$$-2 + (n-2)\lambda = 3n - 8, \qquad (n-2)\lambda = 3n - 6 = 3(n-2).$$

Since $n \geq 3$, we obtain $\lambda = 3$. The determinant agrees: $(-1)^2 3^{n-2} = 3^{n-2}$. Consequently,

$$p_A(t) = (t+1)^2(t-3)^{n-2}, \qquad \mathrm{tr}(A^k) = 2(-1)^k + (n-2)3^k.$$

The last formula follows by powering a diagonal representation and using trace invariance under similarity.

All wrapped up. Victory!

Additional practice

EXERCISE 13.18 FOUNDATION

Consider the real matrix

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

1. Please find its characteristic polynomial and all eigenspaces.
2. Please give real matrices P, D satisfying $A = PDP^{-1}$.
3. Please find an explicit formula for A^k for every integer $k \geq 0$.

WORKED SOLUTION

1. The block structure gives

$$p_A(t) = (t+1)\det \begin{pmatrix} t-4 & -2 \\ -2 & t-4 \end{pmatrix} = (t+1)((t-4)^2 - 4) = (t+1)(t-2)(t-6).$$

The eigenvalues are distinct. Solving the kernel equations, or direct verification, gives

$$E_6 = \mathrm{span}\{(1,1,0)^T\}, \quad E_2 = \mathrm{span}\{(1,-1,0)^T\}, \quad E_{-1} = \mathrm{span}\{(0,0,1)^T\}.$$

Each eigenspace has dimension 1, as also follows from its algebraic multiplicity 1.

2. Place the eigenbasis vectors in the columns:

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \mathrm{diag}(6, 2, -1).$$

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EXERCISE 13.18 FOUNDATION CONTINUED

We have $\det P = -2 \neq 0$, and the eigenvalue equations give $AP = PD$. Thus $A = PDP^{-1}$.

3. Computing PD^kP^{-1} gives

$$A^k = \begin{pmatrix} (6^k + 2^k)/2 & (6^k - 2^k)/2 & 0 \\ (6^k - 2^k)/2 & (6^k + 2^k)/2 & 0 \\ 0 & 0 & (-1)^k \end{pmatrix}.$$

The formula also holds at $k = 0$, giving I_3. For $k \geq 1$, all entries are integers despite the division by 2 in this expression.

We've reached the finish. Victory!

EXERCISE 13.19 CHALLENGE

Let $A \in M_n(\mathbb{F})$ have n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ in $\mathbb{F}$.

1. Please prove that $B \in M_n(\mathbb{F})$ satisfies $AB = BA$ if and only if there is a unique polynomial f of degree at most $n - 1$ (including the zero polynomial) with $B = f(A)$.
2. Over $\mathbb{R}$, please find such a polynomial for

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

WORKED SOLUTION

1. First, every power of A commutes with A, so every linear combination of its powers commutes with it. Thus $B = f(A)$ implies $AB = BA$. Conversely, assume $AB = BA$. Since A has n distinct eigenvalues in the field, it has an eigenbasis $(v_1, \ldots, v_n)$ with $Av_i = \lambda_i v_i$, and every eigenspace is one-dimensional. We have

$$A(Bv_i) = B(Av_i) = \lambda_i Bv_i.$$

Thus $Bv_i \in E_{\lambda_i}(A) = \operatorname{span}\{v_i\}$, including the possibility $Bv_i = 0$. Hence $Bv_i = \mu_i v_i$ for some $\mu_i \in \mathbb{F}$. Define explicitly

$$f(t) = \sum_{i=1}^{n} \mu_i \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

All denominators are nonzero, so this polynomial is defined over $\mathbb{F}$. For $n = 1$, an empty product equals 1. Its degree is at most $n-1$, and at $t = \lambda_i$ only the ith summand remains, giving $f(\lambda_i) = \mu_i$. Therefore

$$f(A)v_i = f(\lambda_i)v_i = \mu_i v_i = Bv_i.$$

The two maps agree on a basis, so $f(A) = B$. If another polynomial g of degree at most $n - 1$ satisfies $g(A) = B$, then $(f - g)(\lambda_i) = 0$ for every i. A nonzero polynomial of degree at most $n - 1$ cannot have n distinct roots, so $f = g$. Uniqueness requires the

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EXERCISE 13.19 CHALLENGE CONTINUED

degree bound; without it, a multiple of p_A may be added.

2. On $(1,1)^T, (1,-1)^T$, the eigenvalues of A are $4, 2$, and those of B are $9, 5$. Writing $f(t) = at + b$ gives

$$4a + b = 9, \qquad 2a + b = 5.$$

Thus $2a = 4$, so $a = 2$, $b = 1$. Indeed,

$$B = 2A + I_2, \qquad f(t) = 2t + 1.$$

Yes! Victory!

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EXERCISE 13.20 CHALLENGE

For $n \geq 2$, define $S(x_1, \ldots, x_n) = (x_n, x_1, \ldots, x_{n-1})$ on $\mathbb{C}^n$.

1. Please find an eigenbasis of S and its characteristic polynomial.
2. For a prime p, please define the same shift on $\mathbb{Z}_p^p$ and prove that it is not diagonalizable over $\mathbb{Z}_p$.

WORKED SOLUTION

Let $\omega = e^{2\pi i/n}$ and define

$$v_k = (1, \omega^{-k}, \omega^{-2k}, \ldots, \omega^{-(n-1)k})^T, \qquad 0 \leq k < n.$$

The first coordinate of Sv_k is $\omega^{-(n-1)k} = \omega^k$. Coordinate $j \geq 2$ is $\omega^{-(j-2)k} = \omega^k \omega^{-(j-1)k}$. Hence $Sv_k = \omega^k v_k$. The n eigenvalues are distinct, so these eigenvectors form a basis, and

$$p_S(t) = \prod_{k=0}^{n-1} (t - \omega^k) = t^n - 1.$$

Over $\mathbb{Z}_p$, $S^p = I$. Since S and I commute and $\binom{p}{k}$ is divisible by p for $1 \leq k < p$, the binomial formula gives

$$(S - I)^p = S^p - I = 0.$$

If S were diagonalizable, so would $S - I$ be. A diagonal matrix with a zero positive power has every diagonal entry zero. This would imply $S = I$, contradicting $S(e_1) = e_2 \neq e_1$ for $p \geq 2$.

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EXERCISE 13.21 CHALLENGE

Let $A \in M_n(\mathbb{C})$ and assume A^2 is diagonalizable. Please prove that A is diagonalizable if and only if $\ker A = \ker A^2$. Please give a 2×2 example in which A^2 is diagonalizable and A is not.

WORKED SOLUTION

If A is diagonalizable, an eigenbasis makes it diagonal. A diagonal entry is zero exactly when its square is zero, giving equality of the kernels. Conversely, suppose the kernels agree. Since A^2 is diagonalizable,

$$\mathbb{C}^n = \bigoplus_{\mu} E_\mu(A^2), \qquad E_\mu(A^2) = \{v : A^2 v = \mu v\}.$$

Each summand is preserved by A, since $A^2(Av) = A(A^2 v) = \mu Av$. On $E_0(A^2) = \ker A$, the map A is zero. For $\mu \neq 0$, choose $\lambda \in \mathbb{C}$ with $\lambda^2 = \mu$. For $v \in E_\mu(A^2)$, set

$$v_+ = \tfrac{1}{2}(v + \lambda^{-1} Av), \qquad v_- = \tfrac{1}{2}(v - \lambda^{-1} Av).$$

Then $v = v_+ + v_-$ and direct substitution gives $Av_+ = \lambda v_+$ and $Av_- = -\lambda v_-$. Their eigenspaces intersect trivially because $\lambda \neq 0$. Thus each summand has a basis of eigenvectors for A, and their union is an eigenbasis of the whole space. For a counterexample, $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $A^2 = 0$. Its only possible eigenvalue is zero. If A were diagonalizable, it would be zero, contradicting its displayed entry.

Victory!

13.5 Jordan Form

DEFINITION 13.7

Let $k \geq 1$ be an integer and $\lambda \in F$. The Jordan block of size k with eigenvalue λ is

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix} = \lambda I_k + N_k.$$

The matrix N_k has 1 immediately above the diagonal and zero elsewhere; set $N_1 = 0$. We have $N_k^k = 0$, while $N_k^{k-1} \neq 0$ for $k \geq 2$. A matrix with a positive power equal to zero is called **nilpotent**.

RECALL AND TOOLS

If the characteristic polynomial of $A \in M_n(F)$ splits into linear factors over F, there is an invertible P such that

$$P^{-1} A P = \operatorname{diag}\big(J_{k_1}(\lambda_1), \ldots, J_{k_s}(\lambda_s)\big).$$

Here diag denotes a block diagonal matrix: the displayed blocks are on the diagonal, with

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zero elsewhere. This is the Jordan form theorem. Over $\mathbb{C}$, the splitting condition always holds. Over $\mathbb{R}$, it holds exactly when all roots of the characteristic polynomial are real. Block sizes and eigenvalues are determined up to the order of the blocks.
Diagonalization occurs exactly when every block has size 1. For a given λ, the sum of its block sizes is its algebraic multiplicity, and the number of its blocks is its geometric multiplicity.

DEFINITION 13.8

A **Jordan chain** for λ is a list $v_1, \dots, v_k$ satisfying

$$v_1 \neq 0, \qquad (A - \lambda I)v_1 = 0, \qquad (A - \lambda I)v_j = v_{j-1} \quad (2 \leq j \leq k).$$

Thus v_1 is an eigenvector; the other chain vectors are called generalized eigenvectors. Every such chain is linearly independent: if $\sum_{j=1}^{k} a_j v_j = 0$ and a_ℓ is the last nonzero coefficient, applying $(A - \lambda I)^{\ell-1}$ gives $a_\ell v_1 = 0$, a contradiction. In this order,

$$A(v_1 \ \cdots \ v_k) = (v_1 \ \cdots \ v_k)J_k(\lambda).$$

If the columns of P form a basis made of these chains, then $AP = PJ$, so $P^{-1}AP = J$.

EXAMPLE 13.3

For

$$A = \begin{pmatrix} 5 & 4 \\ -1 & 1 \end{pmatrix}, \qquad N = A - 3I = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix},$$

we have

$$\chi_A(t) = (t-5)(t-1) + 4 = (t-3)^2, \qquad N^2 = \begin{pmatrix} 4-4 & 8-8 \\ -2+2 & -4+4 \end{pmatrix} = 0.$$

The equation $Nx = 0$ is $x_1 + 2x_2 = 0$, so the eigenspace has dimension one. Choose

$$v_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad Nv_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad Nv_2 = v_1.$$

This is a chain of length 2. With

$$P = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix},$$

we obtain

$$AP = \begin{pmatrix} 6 & 5 \\ -3 & -1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} = J_2(3).$$

The superdiagonal entry 1 records $Av_2 = 3v_2 + v_1$. There is no basis of two eigenvectors.

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EXAMPLE 13.4

Let

$$A = \begin{pmatrix} -1 & 3 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Choose $v_1 = (1,1,0)^T$, $v_2 = (0,1,1)^T$, and $v_3 = (1,0,0)^T$. Direct calculation gives

$$Av_1 = (2,2,0)^T = 2v_1, \qquad Av_2 = (1,3,2)^T = 2v_2 + v_1, \qquad Av_3 = -v_3.$$

The first two vectors form a chain for 2, while the third is an eigenvector for -1. Set

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Then

$$AP = \begin{pmatrix} 2 & 1 & -1 \\ 2 & 3 & 0 \\ 0 & 2 & 0 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Thus $\chi_A(t) = (t-2)^2(t+1)$, but the eigenspace for 2 has dimension one, so A is not diagonalizable.

EXERCISE 13.22

Let $N \in M_7(\mathbb{C})$ satisfy $N^3 = 0$, rank $N = 4$, and rank $N^2 = 2$. Please find its Jordan form and $\dim \ker N^j$ for every integer $j \geq 1$.

WORKED SOLUTION

Every eigenvalue satisfies $\lambda^3 = 0$, so $\lambda = 0$. Each block has size at most 3, since the cube of a larger nilpotent block is nonzero. Let b_k be the number of blocks of size k, for $k = 1, 2, 3$. For each $J_k(0)$,

$$\operatorname{rank} J_k(0) = k - 1, \qquad \operatorname{rank}(J_k(0)^2) = \max\{k-2, 0\}.$$

These are the numbers of nonzero columns: the first power moves one step along the chain, and the second moves two steps. Hence

$$\begin{aligned} b_1 + 2b_2 + 3b_3 &= 7, \\ b_2 + 2b_3 &= 4, \\ b_3 &= 2. \end{aligned}$$

The last equation gives $b_3 = 2$, the second gives $b_2 = 0$, and the first gives $b_1 = 1$. The Jordan form is therefore

$$\operatorname{diag}\big(J_3(0), J_3(0), J_1(0)\big).$$

Rank-nullity gives

$$\dim \ker N = 7 - 4 = 3, \qquad \dim \ker N^2 = 7 - 2 = 5.$$

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EXERCISE 13.22 CONTINUED

For $j \geq 3$, $N^j = 0$, so $\dim \ker N^j = 7$.

And we're done. Victory!

EXERCISE 13.23

Let $A \in M_n(\mathbb{C})$ satisfy $A^3 = A^2$ and $\operatorname{rank} A = \operatorname{rank} A^2$. Please prove that $A^2 = A$. Please construct an example with $A^3 = A^2$ but $A^2 \neq A$.

WORKED SOLUTION

Every eigenvalue satisfies $\lambda^3 = \lambda^2$, so $\lambda \in \{0, 1\}$. Pass to Jordan form. On a block for 0, write the block as N. The condition is

$$N^3 = N^2, \qquad N^2(N - I) = 0.$$

The matrix $N - I$ is invertible: if $N^k = 0$, then $(I - N)^{-1} = I + N + \cdots + N^{k-1}$. Thus $N^2 = 0$, and the block has size at most 2. On a block for 1, write $J = I + N$. Then

$$J^3 - J^2 = N + 2N^2 + N^3 = N(I + N)^2 = 0.$$

Also $I + N$ is invertible, with inverse $I - N + \cdots + (-N)^{k-1}$, so $N = 0$. Every block for 1 therefore has size 1. A block $J_2(0)$ contributes 1 to rank A and zero to rank A^2. Size-one blocks for $0, 1$ contribute the same rank to both. Equality of the ranks excludes every $J_2(0)$. Thus $A = PDP^{-1}$ for a diagonal matrix D with entries $0, 1$, so $A^2 = PD^2P^{-1} = PDP^{-1} = A$. For example, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ satisfies $B^3 = B^2 = 0$, but $B^2 \neq B$. Here $\operatorname{rank} B = 1$ and $\operatorname{rank} B^2 = 0$.

You know what that means? V-I-C-T-O-R-Y!

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EXERCISE 13.24 CHALLENGE

Let $\dim V = n$ over a field F, and let $T : V \to V$ be linear with $T^2 = 0$. Put $r = \operatorname{rank} T$. Please prove that $2r \le n$ and that some basis gives the matrix

$$\begin{pmatrix} 0 & I_r & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The three row and column block sizes are $r, r, n - 2r$. Blocks of size zero are omitted.

WORKED SOLUTION

The identity $T^2 = 0$ gives $\operatorname{Im} T \subseteq \ker T$. Rank-nullity yields $r \le n - r$, so $2r \le n$. Choose a basis $u_1, \dots, u_r$ of the image and extend it to a basis $u_1, \dots, u_r, z_1, \dots, z_{n-2r}$ of the kernel. For each i, choose w_i with $T(w_i) = u_i$. The list

$$\mathcal{B} = (u_1, \dots, u_r, w_1, \dots, w_r, z_1, \dots, z_{n-2r})$$

is independent. Applying T to a zero linear combination shows that all coefficients of the w_i vanish, because the u_i are independent. The remaining combination is in a basis of the kernel, so all its coefficients vanish too. The list has n vectors and is therefore a basis. In this basis $T(u_i) = T(z_j) = 0$ and $T(w_i) = u_i$. Writing these images as columns gives the displayed matrix. When $r = 0$, $T = 0$ and every basis works.

Another one down. Victory!

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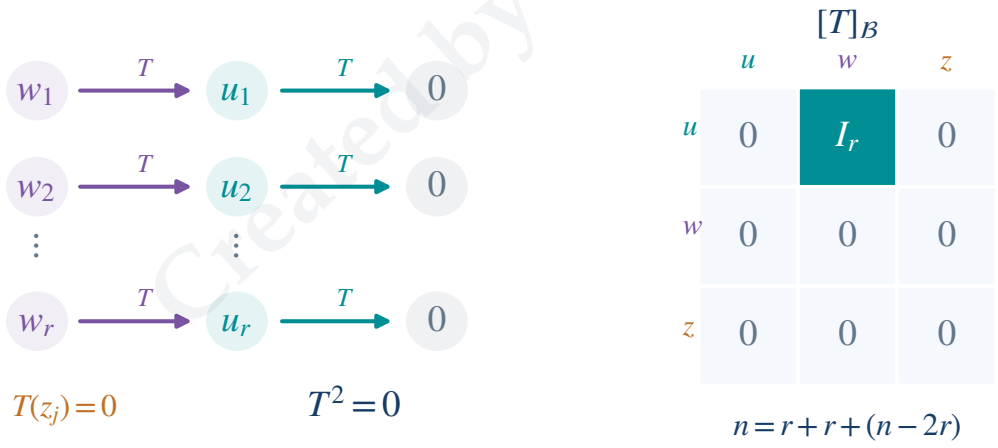


Figure 13.1. The schematic shows the case $r \ge 3$. Each w_i maps to u_i, and each u_i maps to zero. The vectors z_j complete $u_1, \dots, u_r$ to a basis of the kernel. The basis groups are ordered u, w, z.

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EXERCISE 13.25 CHALLENGE

Let $A \in M_n(F)$ have rank 1, where $n \geq 2$, and put $\alpha = \operatorname{tr} A$, the sum of the diagonal entries. Please prove that A is similar to

$$\begin{cases} \operatorname{diag}(\alpha, 0, \ldots, 0), & \alpha \neq 0, \\ E_{12}, & \alpha = 0, \end{cases}$$

where E_{12} has a single 1 in position $(1, 2)$. Please deduce that two rank-one matrices of the same size and trace are similar.

WORKED SOLUTION

Write $A = uv^T$ with $u, v \neq 0$ and define $\varphi(x) = v^T x$. Then

$$Ax = \varphi(x)u, \qquad \operatorname{tr} A = \sum_i u_i v_i = \varphi(u) = \alpha.$$

Since $\varphi \neq 0$, its kernel has dimension $n-1$. If $\alpha \neq 0$, then $u \notin \ker \varphi$. A basis $\mathcal{B} = (u, z_2, \ldots, z_n)$ obtained by adjoining u to a basis of the kernel satisfies $Au = \alpha u$ and $Az_j = 0$. Its representing matrix is $\operatorname{diag}(\alpha, 0, \ldots, 0)$. If $\alpha = 0$, then $u \in \ker \varphi$. Choose w with $\varphi(w) = 1$ and extend u to a basis $(u, z_3, \ldots, z_n)$ of the kernel. The list $\mathcal{B} = (u, w, z_3, \ldots, z_n)$ is a basis: applying φ to a zero combination forces the coefficient of w to vanish; the remaining coefficients vanish by independence in the kernel. Now $Au = Az_j = 0$ and $Aw = u$, giving the matrix E_{12}. Two rank-one matrices with the same size and trace are therefore similar to the same displayed matrix, hence to one another.

We've got it. Victory!

13.6 Inner Product Spaces

RECALL AND TOOLS

Throughout this chapter, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. Write $\bar{\cdot} : \mathbb{F} \to \mathbb{F}$ for standard conjugation: that is, if $\mathbb{F} = \mathbb{C}$, then $\overline{a + ib} = a - ib$; if $\mathbb{F} = \mathbb{R}$, then $\overline{x} = x$ for every $x \in \mathbb{R}$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An *inner product* on V is a map

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

satisfying the following axioms for all $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{F}$.

1. **Linearity in the first argument:**

$$\langle \alpha u + \beta v,\ w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle.$$

2. **Conjugate symmetry (Hermitian symmetry):**

$$\langle u, v \rangle = \overline{\langle v, u \rangle}.$$

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3. **Positive definiteness:**

$$\langle v, v\rangle \in \mathbb{R}, \qquad \langle v, v\rangle \geq 0, \qquad \langle v, v\rangle = 0 \iff v = 0.$$

Formal observation: Linearity in the first argument and conjugate symmetry imply *conjugate linearity in the second argument*: for all $\alpha, \beta \in \mathbb{F}$ and all $u, v, w \in V$,

$$\langle u,\ \alpha v + \beta w\rangle = \overline{\alpha}\,\langle u, v\rangle + \overline{\beta}\,\langle u, w\rangle.$$

When $\mathbb{F} = \mathbb{R}$, this gives ordinary linearity in the second argument as well.

EXAMPLE 13.5

Standard examples of inner products:

- On $\mathbb{C}^n$ (column vectors $u, v \in \mathbb{C}^n$):

$$\langle u, v\rangle := u^T \overline{v}.$$

  On $\mathbb{R}^n$:

$$\langle u, v\rangle := u^T v.$$

- On $M_{m\times n}(\mathbb{C})$:

$$\langle A, B\rangle := \operatorname{tr}\big(A^T \overline{B}\big).$$

- On $\mathbb{R}_n[x]$ (or on $C([a, b], \mathbb{R})$), where $a < b$:

$$\langle p, q\rangle := \int_a^b p(x)\, q(x)\, dx.$$

DEFINITION 13.9

Let $(V, \langle\cdot,\cdot\rangle)$ be a vector space equipped with an inner product. The **norm** induced by the inner product is defined, for each $v \in V$, by

$$\|v\| := \sqrt{\langle v, v\rangle}.$$

Thus $\|v\| \in [0, \infty)$; in particular, $\|v\| = 0$ if and only if $v = 0$.

DEFINITION 13.10

Inner product space: If V is a vector space over $\mathbb{F}$ and $\langle\cdot,\cdot\rangle$ is an inner product on V, then the pair

$$\big(V, \langle\cdot,\cdot\rangle\big)$$

is called an **inner product space**.

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PROPOSITION 13.5

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For all $n, m \in \mathbb{N}$, all $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m \in \mathbb{F}$, and all $v_1, \ldots, v_n, u_1, \ldots, u_m \in V$,

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \overline{\beta_j}\, \langle v_i, u_j \rangle.$$

When $\mathbb{F} = \mathbb{R}$, we have $\overline{\beta_j} = \beta_j$, so the formula reduces to

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \beta_j\, \langle v_i, u_j \rangle.$$

13.7 Basic Exercises and Results

EXERCISE 13.26

Let $n \geq 1$. On $V = M_n(\mathbb{R})$, define

$$f(A, B) := \operatorname{tr}(AB).$$

Please determine for which values of n this is an inner product, and

WORKED SOLUTION

If $n = 1$, identify matrices with real numbers. Then $f(a, b) = ab$ is the standard inner product on $\mathbb{R}$. If $n \geq 2$, choose $A = E_{21}$: that is, $a_{21} = 1$ and all other entries are zero. Then $A \neq 0$, but $A^2 = 0$, so

$$f(A, A) = \operatorname{tr}(A^2) = 0.$$

Positive definiteness therefore fails, and the function is not an inner product when $n \geq 2$.

That does it. Victory!

EXERCISE 13.27

Let $n \in \mathbb{N}$, and let $M \in M_n(\mathbb{R})$ be symmetric and positive definite; that is,

$$M^T = M, \qquad x^T M x > 0 \ \ \forall x \in \mathbb{R}^n \setminus \{0\}.$$

Define $\langle\cdot,\cdot\rangle_M : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle x, y \rangle_M := x^T M y.$$

1. Please prove that $\langle\cdot,\cdot\rangle_M$ is an inner product on $\mathbb{R}^n$.
2. Please show that the induced norm is $\|x\|_M = \sqrt{x^T M x}$.
3. Physical example: suppose a linear mechanical system is described by a mass matrix M (in some system of generalized coordinates). Please show that its kinetic energy

$$T(v) = \frac{1}{2}\, v^T M v$$

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EXERCISE 13.27 CONTINUED

equals $\frac{1}{2}\|v\|_M^2$.

WORKED SOLUTION

1. We verify the inner product axioms.

   Linearity in the first argument: Let $x_1, x_2, y \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then

   $$\langle \alpha x_1 + \beta x_2,\, y\rangle_M = (\alpha x_1 + \beta x_2)^T M y = \alpha x_1^T M y + \beta x_2^T M y = \alpha\langle x_1, y\rangle_M + \beta\langle x_2, y\rangle_M.$$

   Symmetry: For every $x, y \in \mathbb{R}^n$,

   $$\langle x, y\rangle_M = x^T M y.$$

   Since this is a real scalar, $(x^T My)^T = x^T My$, so

   $$\langle x, y\rangle_M = (x^T My)^T = y^T M^T x = y^T M x = \langle y, x\rangle_M,$$

   where we used $M^T = M$.

   Positive definiteness: For every $x \in \mathbb{R}^n$,

   $$\langle x, x\rangle_M = x^T M x \geq 0,$$

   and $\langle x, x\rangle_M = 0$ if and only if $x = 0$, because M is positive definite.

   Hence $\langle \cdot, \cdot\rangle_M$ is an inner product on $\mathbb{R}^n$.

2. By the definition of the norm induced by an inner product, for every $x \in \mathbb{R}^n$,

   $$\|x\|_M := \sqrt{\langle x, x\rangle_M} = \sqrt{x^T M x}.$$

3. For every $v \in \mathbb{R}^n$,

   $$T(v) = \frac{1}{2} v^T M v = \frac{1}{2}\langle v, v\rangle_M = \frac{1}{2}\|v\|_M^2,$$

   so the kinetic energy is half the square of the norm induced by $\langle \cdot, \cdot\rangle_M$.

All wrapped up. Victory!

EXERCISE 13.28

Let $(V, \langle \cdot, \cdot\rangle)$ be an inner product space over $\mathbb{C}$, and let $u, v \in V$ satisfy $u \neq v$. Please show that there is $w \in V$ such that $\langle w, u\rangle \neq \langle w, v\rangle$.

WORKED SOLUTION

Choose $w = u - v$. Since $u \neq v$, the vector w is nonzero. Conjugate linearity in the second argument implies additivity, so

$$\langle w, u\rangle - \langle w, v\rangle = \langle w, u - v\rangle = \langle u - v, u - v\rangle = \|u - v\|^2 > 0.$$

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EXERCISE 13.28 CONTINUED

Therefore $\langle w, u\rangle \neq \langle w, v\rangle$.

We've reached the finish. Victory!

13.8 Norms and the Cauchy-Schwarz Inequality

DEFINITION 13.11

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. A **norm** on V is a function

$$\|\cdot\| : V \to [0, \infty)$$

satisfying the following axioms for all $u, v \in V$ and all $\alpha \in \mathbb{F}$:

1. **Positive definiteness:**
$$\|v\| \geq 0, \qquad \|v\| = 0 \iff v = 0.$$
2. **Homogeneity:**
$$\|\alpha v\| = |\alpha|\,\|v\|.$$
3. **Triangle inequality:**
$$\|u + v\| \leq \|u\| + \|v\|.$$

Here $|\alpha|$ denotes absolute value over $\mathbb{R}$ or modulus over $\mathbb{C}$. A vector space equipped with a norm is a **normed space**, denoted $(V, \|\cdot\|)$.

PROPOSITION 13.6

Cauchy-Schwarz inequality. For all $u, v \in V$,

$$|\langle u, v\rangle| \leq \|u\|\,\|v\|.$$

Equality holds if and only if u, v are linearly dependent: that is, there is $\alpha \in \mathbb{F}$ such that $u = \alpha v$ or $v = \alpha u$.

EXERCISE 13.29

Let $\omega, \zeta, \xi > 0$. Please prove the inequality

$$\frac{\omega}{\zeta + \xi} + \frac{\zeta}{\xi + \omega} + \frac{\xi}{\omega + \zeta} \geq \frac{(\omega + \zeta + \xi)^2}{2(\omega\zeta + \zeta\xi + \xi\omega)}.$$

WORKED SOLUTION

Use the standard inner product on $\mathbb{R}^3$, namely $\langle x, y\rangle = x^T y$.
Define the vectors

$$x = \left(\sqrt{\frac{\omega}{\zeta + \xi}}, \sqrt{\frac{\zeta}{\xi + \omega}}, \sqrt{\frac{\xi}{\omega + \zeta}}\right), \qquad y = \left(\sqrt{\omega(\zeta + \xi)}, \sqrt{\zeta(\xi + \omega)}, \sqrt{\xi(\omega + \zeta)}\right).$$

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EXERCISE 13.29 CONTINUED

Then

$$\langle x,y\rangle = \sqrt{\frac{\omega}{\zeta+\xi}}\sqrt{\omega(\zeta+\xi)} + \sqrt{\frac{\zeta}{\xi+\omega}}\sqrt{\zeta(\xi+\omega)} + \sqrt{\frac{\xi}{\omega+\zeta}}\sqrt{\xi(\omega+\zeta)} = \omega+\zeta+\xi.$$

Also,

$$\|x\|^2 = \langle x,x\rangle = \left(\sqrt{\frac{\omega}{\zeta+\xi}}\right)^2 + \left(\sqrt{\frac{\zeta}{\xi+\omega}}\right)^2 + \left(\sqrt{\frac{\xi}{\omega+\zeta}}\right)^2 = \frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta},$$

and

$$\begin{aligned}\|y\|^2 &= \langle y,y\rangle \\ &= \left(\sqrt{\omega(\zeta+\xi)}\right)^2 + \left(\sqrt{\zeta(\xi+\omega)}\right)^2 + \left(\sqrt{\xi(\omega+\zeta)}\right)^2 \\ &= \omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta).\end{aligned}$$

Expand the last sum:

$$\omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta) = (\omega\zeta+\omega\xi) + (\zeta\xi+\zeta\omega) + (\xi\omega+\xi\zeta) = 2(\omega\zeta+\zeta\xi+\xi\omega).$$

The Cauchy-Schwarz inequality gives

$$\langle x,y\rangle^2 \le \|x\|^2\,\|y\|^2,$$

and hence

$$(\omega+\zeta+\xi)^2 \le \left(\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta}\right)\cdot 2(\omega\zeta+\zeta\xi+\xi\omega).$$

Since $2(\omega\zeta+\zeta\xi+\xi\omega) > 0$, division yields

$$\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta} \ge \frac{(\omega+\zeta+\xi)^2}{2(\omega\zeta+\zeta\xi+\xi\omega)},$$

Yes! Victory!

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EXERCISE 13.30 CHALLENGE

Let V be a complex inner product space, with the inner product linear in its first argument. Please prove that

$$4\langle u, v\rangle = \|u+v\|^2 - \|u-v\|^2 + i\big(\|u+iv\|^2 - \|u-iv\|^2\big).$$

Let $T : V \to V$ be complex-linear and satisfy $\|Tx\| = \|x\|$ for every x. Please deduce that $\langle Tu, Tv\rangle = \langle u, v\rangle$ for every u, v.

WORKED SOLUTION

Put $z = \langle u, v\rangle$. Expanding gives

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2 + z + \overline{z}, \qquad \|u-v\|^2 = \|u\|^2 + \|v\|^2 - z - \overline{z}.$$

Their difference is $4\,\mathrm{Re}\, z$. Conjugate-linearity in the second argument gives $\langle u, iv\rangle = -iz$ and $\langle iv, u\rangle = i\overline{z}$. Consequently

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 - iz + i\overline{z},$$
$$\|u-iv\|^2 = \|u\|^2 + \|v\|^2 + iz - i\overline{z}.$$

Their difference is $4\,\mathrm{Im}\, z$, proving the formula. Apply it to Tu, Tv. Complex linearity gives $Tu \pm Tv = T(u \pm v)$ and $Tu \pm iTv = T(u \pm iv)$. Norm preservation replaces all four norms by the original ones, so the inner product is preserved as well.

V-I-C-T-O-R-Y!

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